0. Introduction

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FS26 MTH345: Deep Generative Models: A Statistical Perspective

ULSAN NATIONAL INSTITUTE OF SCIENCE AND TECHNOLOGY

Lab Homepage

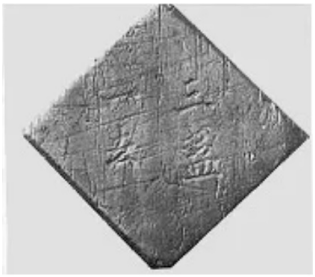
Outline

1. Generative Model

2. Course Description

Motivating Example



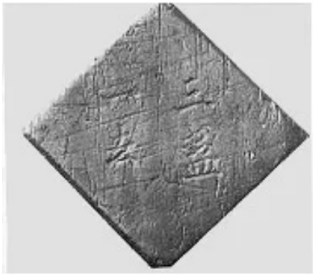


- This course aims to cover the mathematical foundations of **generative models**, extending to early deep learning approaches.
- As a motivating example, we consider learning and generating outcomes from the unique die shown above.

Left: A replica at the Gyeongju National Museum. Right: Inscriptions on each face recorded at the time of excavation.

Motivating Example





- *Juryeonggu* (酒令具; AD668 $\sim$ AD935) is an ancient die used for drinking games during a historical dynasty in what is now South Korea.
- It has 14 faces, six square (□) and eight hexagonal (⎔), with each face assigning a penalty, e.g., 三盞一去: Drink three cups at once.
- **Q:** How can we model the stochastic nature of this die, and computationally generate simulated outcomes based on it?

Motivating Example

- *Probabilistic models* serve an essential role in describing the uncertainty of events arising from such random experiments.
- For example, we may introduce $\boldsymbol{p} = (p_1, \ldots, p_{14})^T \in \{\boldsymbol{p} : p_i \geq 0, \sum_{i=1}^{14} p_i = 1\}$ to represent (or *parameterize*) the probability of each face:

Outcome	1st □	2nd □	3rd □	4th □	5th □	6th □	1st ⬡
Probability	p_1	p_2	p_3	p_4	p_5	p_6	p_7

Outcome	2nd ⬡	3rd ⬡	4th ⬡	5th ⬡	6th ⬡	7th ⬡	8th ⬡
Probability	p_8	p_9	p_{10}	p_{11}	p_{12}	p_{13}	p_{14}

- With $\boldsymbol{p}$, the probability of observing any of square faces can be expressed as:

$$p_1 + \cdots + p_6 \tag{1}$$

Motivating Example

- When $\boldsymbol{p} = (1/14, \ldots, 1/14)^T$, it corresponds to the uniform model, where all faces are equally likely to occur:

Outcome	1st □	2nd □	3rd □	4th □	5th □	6th □	1st ⬡
Probability	1/14	1/14	1/14	1/14	1/14	1/14	1/14

Outcome	2nd ⬡	3rd ⬡	4th ⬡	5th ⬡	6th ⬡	7th ⬡	8th ⬡
Probability	1/14	1/14	1/14	1/14	1/14	1/14	1/14

- The probability of observing any of the square faces is:

$$p_1 + \cdots + p_6 = \frac{6}{14}. \tag{2}$$

- In a similar manner, we can express the probability of any event using this probability table parameterized by $\boldsymbol{p}$.

Motivating Example

- When $\boldsymbol{p} = (\underbrace{1/12, \ldots, 1/12}_{6}, \underbrace{1/16, \ldots, 1/16}_{8})^T$, it yields another probability table:

Outcome	1st □	2nd □	3rd □	4th □	5th □	6th □	1st ⬡
Probability	1/12	1/12	1/12	1/12	1/12	1/12	1/16

Outcome	2nd ⬡	3rd ⬡	4th ⬡	5th ⬡	6th ⬡	7th ⬡	8th ⬡
Probability	1/16	1/16	1/16	1/16	1/16	1/16	1/16

- The probability of observing any of the square faces is now:

$$p_1 + \cdots + p_6 = \frac{1}{2}. \tag{3}$$

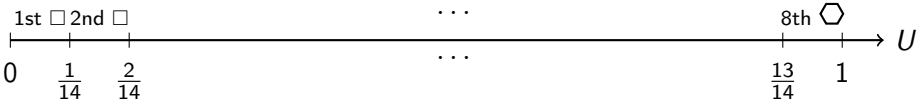
- Using the parameter vector $\boldsymbol{p}$, we can characterize all uncertain events regarding the die (i.e., its *stochasticity*), e.g.,

$$P(\text{observing three consecutive 1st } \square) = p_1^3. \tag{4}$$

Motivating Example

- For a given $\boldsymbol{p}$, we can use the corresponding probabilistic model to **generate** synthetic samples.
- For example, when $\boldsymbol{p} = (1/14, \ldots, 1/14)^T$, we can sample a number U from $[0, 1]$ randomly, and determine the outcome as follows:

$$\text{Outcome} = \begin{cases} \text{1st } \square & \text{if} \quad 0 \leq U \leq 1/14 \\ \text{2nd } \square & \text{if} \quad 1/14 < U \leq 2/14 \\ \quad \vdots & \quad \vdots \\ \text{8th } \hexagon & \text{if} \quad 13/14 < U \leq 1 \end{cases}$$



Motivating Example

- Let's say that we sample the following 5 numbers:

$$3421, .7812, .0534, .9123, .4456 \tag{5}$$

- Based on the previous rule, the corresponding synthetic samples are:

$$5\text{th } \square, 5\text{th } \hexagon, 1\text{st } \square, 7\text{th } \hexagon, 1\text{st } \hexagon \tag{6}$$

  Intuitively, as we repeat this process infinitely many times, the (empirical) probability table of these synthetic samples will converge to the one defined by $\boldsymbol{p}$.

- **Q**: Each $\boldsymbol{p}$ defines a distinct probability table, generating a different synthetic dataset. What would be the optimal choice of $\boldsymbol{p}$ to best mimic the real data?

Motivating Example

- The *(density) estimation* is the process of finding the optimal probability table based on *data*.
- Wang et al. (2005) conducted an experiment rolling the die $n = 7,000$ times using two different materials: wood and Fiberglass Reinforced Plastic (FRP).

Outcome	1st □	2nd □	3rd □	4th □	5th □	6th □	1st ⬡
Relative Frequency	$\frac{523}{7000}$	$\frac{533}{7000}$	$\frac{477}{7000}$	$\frac{525}{7000}$	$\frac{518}{7000}$	$\frac{492}{7000}$	$\frac{456}{7000}$

Outcome	2nd ⬡	3rd ⬡	4th ⬡	5th ⬡	6th ⬡	7th ⬡	8th ⬡
Relative Frequency	$\frac{512}{7000}$	$\frac{506}{7000}$	$\frac{488}{7000}$	$\frac{505}{7000}$	$\frac{487}{7000}$	$\frac{482}{7000}$	$\frac{496}{7000}$

- The vector $\hat{\boldsymbol{p}} = \left(\frac{523}{7000}, \ldots, \frac{496}{7000}\right)^T$ [1] yields the above probability table.

[1] A symbol with a hat (e.g., $\hat{p}$) indicates an estimate of the corresponding parameter.

Deep Generative Model



- The previous example illustrates the process of constructing probabilistic models, estimating the optimal models, and generating synthetic samples based on them.
- However, the high-dimensional and complex structures of recent data necessitate advanced statistical methods.
- Recent models have enabled the learning and generation of such complex data. They are now commonly referred to as *Generative AI* or *Deep Generative Models*.

Images are from the CelebA dataset (Liu et al., 2015).

Deep Generative Model



- Let $\boldsymbol{X} = (X_1, \ldots, X_d)^T$ be an observation vector following $p(\boldsymbol{x})$, where, for example,

$$d = 1,024 \text{ (width)} \times 1,024 \text{ (height)} \times 3 \text{ (RGB color channels)} \approx 3.15\text{M} \quad (7)$$

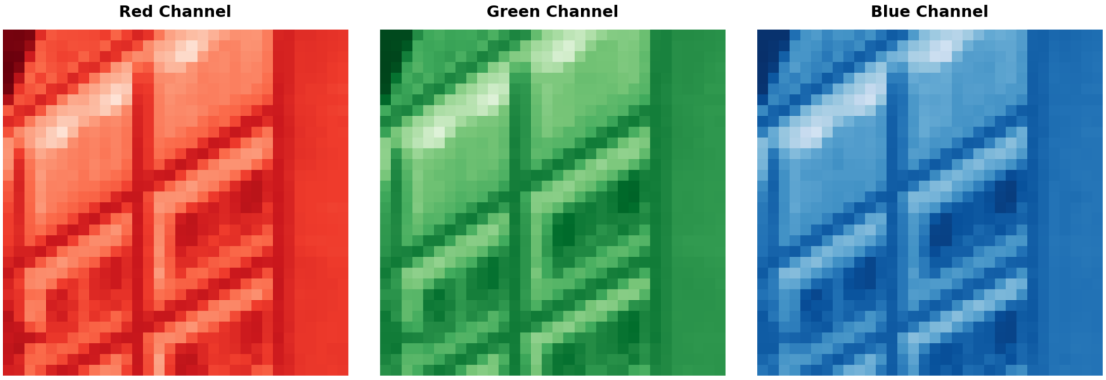
  for the color images shown above.
- Each pixel color value ranges from 0 to 255, comprising a total of 256 discrete integer values.

Deep Generative Model



Source: Park (2022); Photograph by Kyung-chae Kim.

Deep Generative Model

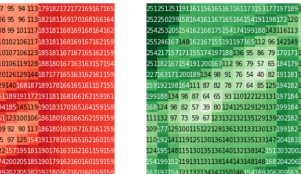


Deep Generative Model

Red Channel



Green Channel



Blue Channel



Deep Generative Model

- Deep generative models learn the *distribution* (i.e., probability table) of $\boldsymbol{X}$:

$$P_{\text{data}}(X_1 = x_1, \ldots, X_d = x_d). \tag{8}$$

- With $P_{\text{data}}(\boldsymbol{x})$, we can augment data, investigate cluster structures, and detect distinct patterns in data.

Application: Image Generation



- Generative models can synthesize photo-realistic data without prior information such as age and race in face generation.
- These models can be used to augment data, generate privacy-free samples, and enhance virtual reality experiences.

The generated image is from `http://thispersondoesnotexist.com/`

Application: Text-to-Image Generation



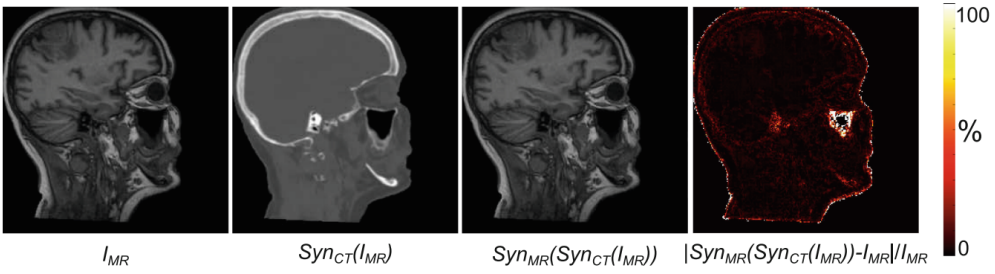
- The images are generated based on the following text description:

  "There is a clean desk in the middle. Outside the window, a whale shark is swimming in the dark night sky above Manhattan."

- The generated images reflect the semantic information provided in the text.

Images are generated with DALL-E-2 (Ramesh et al., 2022).

Application: Cross Modality Transfer



- Generative models are also useful in imputing missing modalities, e.g., magnetic resonance (MR) $\rightarrow$ computed tomography (CT), or enhancing data resolution.

Images are from Wolterink et al. (2017).

Challenges

- For data such as 4K-resolution color images, the dimensionality is $d = 3,840 \times 2,160 \times 3 \approx 24\text{M}$.
- When we introduce parameters for every possible configuration of pixel values, e.g.,

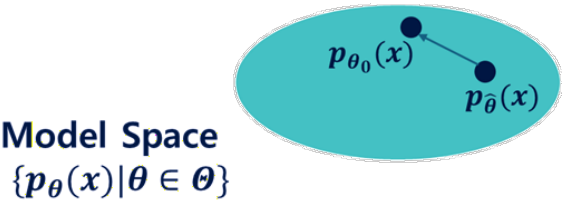
$$p_{0,0,\ldots,0} := P(X_1 = 0, X_2 = 0, \ldots, X_d = 0), \tag{9}$$

  we can express any possible probability table. However, the number of parameters is astronomically large, approximately

$$256^{3,840\times2,160\times3} \approx 10^{60\text{M}}. \tag{10}$$

- Furthermore, image, video, audio, and language data exhibit complex structures, so we need both high-performance and interpretable models.

From a Statistical Perspective



- We formalize the process illustrated in the example as follows.
- Let X (or $\boldsymbol{X}$) be our target of interest. We denote the parameter by θ (e.g., $\boldsymbol{p}$) and define the probability mapping (i.e., the probability table) by:

$$p_\theta(x) := P_\theta(X = x), \tag{11}$$

where P_θ represents the probability distribution parameterized by θ.

Outline

1. Generative Model

2. Course Description

Course Description

Undergraduate				Graduate
Freshman	*Sophomore*	*Junior*	*Senior*	
Basics •Calculus •AI Programming	**Related Math** •Linear Algebra •Statistics •Mathematical Analysis	**This Course** •Deep Generative Models: A Statistical Perspective		**Advanced Course** • AI518: Deep Generative Models

Lecture Topics

M1. Foundations of Probabilistic Modelling (Weeks 01–07)

1. Fundamentals of Probability Theory
2. Random Variables
3. Likelihood Maximization Principle
4. Basics of Probabilistic Modelling

M2. Likelihood Ratio-based Generative Models (Weeks 08–13)

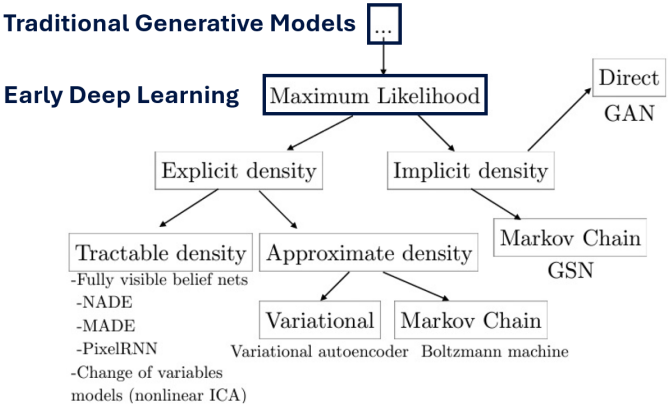
5. Autoregressive Model
6. Energy-based Model
7. Variational Autoencoder
8. Generative Adversarial Network

M3. Generative Models with Other Statistical Distance (Weeks 13–15)

9. Integral Probability Metric-based Approaches
10. Wasserstein Distance-based Approaches
11. Score-based Approaches

Lecture Topics

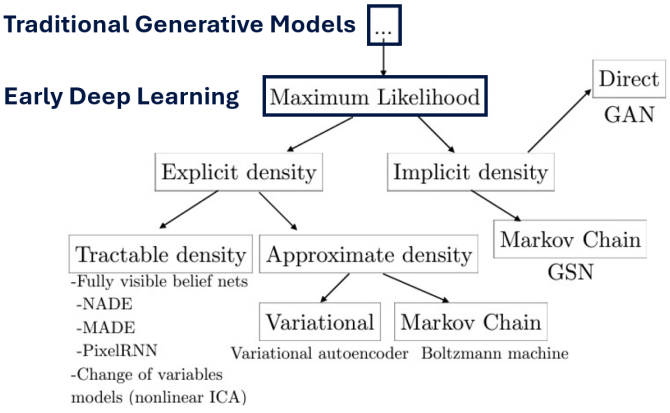
- **M1. Foundations of Probabilistic Modelling:** Core foundations for probabilistic and generative modelling (Probability theory, MLE, Bayesian networks, PCA/ICA).



The figure is edited from Goodfellow (2016).

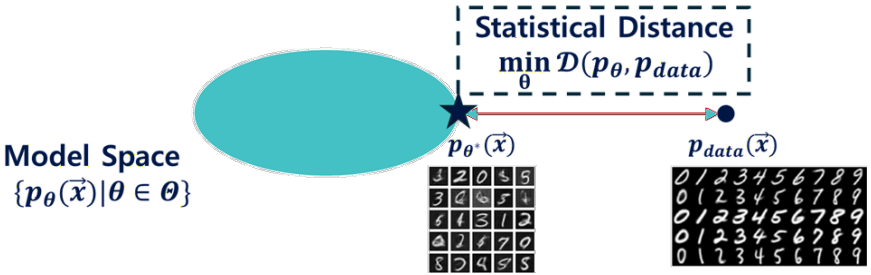
Lecture Topics

- **M2. Likelihood Ratio-based Generative Models:** Representative likelihood ratio-based models prevalent through 2014 (Autoregressive, EBM, VAE, and GAN).

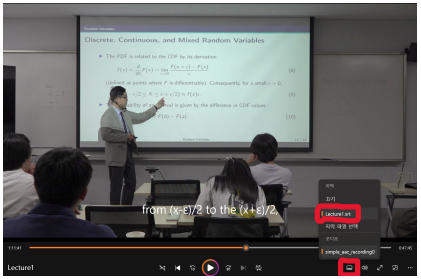


Lecture Topics

- **M3. Generative Models with Other Statistical Distance:** Intuitive principles of measuring distances between distributions, and early generative models applying these specific statistical distances.



Acknowledgements & Pilot Course



- The development of this course was supported by UNIST.
- A pilot version of this course was previously conducted. Lecture recordings with English subtitles and lecture notes are available at:
  `https://github.com/kyg0910/DeepGenerativeModel-A-Statistical-Perspective`

References I

Goodfellow, I. (2016). Nips 2016 tutorial: Generative adversarial networks. *arXiv preprint arXiv:1701.00160*.

Liu, Z., Luo, P., Wang, X., and Tang, X. (2015). Deep learning face attributes in the wild. In *Proceedings of International Conference on Computer Vision (ICCV)*.

Park, T.-j. (2022). UNIST News [Photo Essay] Writer Kim's 'Cherry Blossom Ending'. `https://admu-intl.unist.ac.kr/unist/center/news-en.do`. Accessed: 2026-04-23.

Ramesh, A., Dhariwal, P., Nichol, A., Chu, C., and Chen, M. (2022). Hierarchical text-conditional image generation with clip latents. *arXiv preprint arXiv:2204.06125*, 1(2):3.

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References II

Wolterink, J. M., Dinkla, A. M., Savenije, M. H., Seevinck, P. R., van den Berg, C. A., and Išgum, I. (2017). Deep mr to ct synthesis using unpaired data. In *International workshop on simulation and synthesis in medical imaging*, pages 14–23. Springer.